Que-1 Install MySQL Workbench on your computer, connect to the local MySQL server, and take a screenshot of the default databases shown in the Schemas panel.

Ans-

Steps:-

1. Open MySQL Workbench.
2. Connect to your local MySQL server (usually `Local instance MySQL80`).
3. On the left side, locate the Schemas panel.
4. Expand the schemas list.
5. Take a screenshot:
   - Windows: `Win + Shift + S`
   - Mac: `Command + Shift + 4` (select area) or `Command + Shift + 3` (full screen)
6. Save the screenshot and submit it.

Que-2 Create a new database called InstaClone in MySQL Workbench, then create a table named Users with columns: user_id (INT, primary key), username (VARCHAR), email (VARCHAR), and followers_count (INT).

Ans-

```
CREATE DATABASE InstaClone;

USE InstaClone;

CREATE TABLE Users (
   user_id INT PRIMARY KEY,
   username VARCHAR(50),
   email VARCHAR(100),
   followers_count INT
);
```

Que-3 Insert 3 sample users into the Users table you created, using realistic Instagram-style usernames and follower counts.

Ans-

```
INSERT INTO Users (user_id, username, email, followers_count)
VALUES
(1, 'travel_with_hasib', 'hasib@gmail.com', 1250),
(2, 'foodie_diaries', 'foodie@gmail.com', 5600),
(3, 'tech_guru_2026', 'techguru@gmail.com', 8900);
```

Que-4 Create another table in the InstaClone database called Posts with columns: post_id (INT, primary key), user_id (INT), caption (VARCHAR), and post_date (DATE). Add a foreign key from Posts.user_id to Users.user_id to establish a relationship.
Ans-

```
USE InstaClone;

CREATE TABLE Posts (
   post_id INT PRIMARY KEY,
   user_id INT,
   caption VARCHAR(255),
   post_date DATE,
   FOREIGN KEY (user_id) REFERENCES Users(user_id)
);
```

Que-5 Write a short comparison (3-4 lines each) of MySQL, PostgreSQL, Oracle, and SQLite, focusing on where each one is commonly used (for example: web apps, mobile apps, enterprise systems, etc.).
Ans-

You can write the answer like this in your assignment:

MySQL

MySQL is a popular open-source relational database management system. It is widely used in web applications such as blogs, e-commerce websites, and content management systems. MySQL is known for its speed, ease of use, and strong community support. It is commonly used with PHP, Python, and Java applications.

PostgreSQL

PostgreSQL is an advanced open-source database known for reliability and powerful features. It is commonly used in enterprise applications, data analytics, and geographic information systems (GIS). PostgreSQL supports complex queries, JSON data, and custom data types. It is preferred when applications require high performance and data integrity.

Oracle

Oracle Database is a commercial database system designed for large-scale enterprise environments. It is widely used by banks, government organizations, healthcare systems, and large corporations. Oracle provides excellent security, scalability, and high availability features. It is suitable for mission-critical applications that handle large amounts of data.

SQLite

SQLite is a lightweight, serverless database that stores data in a single file. It is commonly used in mobile applications, desktop software, and embedded systems. SQLite requires minimal setup and does not need a separate database server. It is ideal for small to medium-sized applications with local data storage needs.

Comparison Table

Database	Common Use Cases
MySQL	Web applications, e-commerce websites, CMS
PostgreSQL	Enterprise apps, analytics, GIS systems
Oracle	Banking, healthcare, government, large enterprises
SQLite	Mobile apps, desktop apps, embedded systems